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import cv2
import urllib.request
from google.colab.patches import cv2_imshow

# Download a sample video automatically
# Previous URLs resulted in 404, 403, or SSL errors. Trying a common example video from w3schools.
video_url = "https://www.w3schools.com/html/mov_bbb.mp4"
urllib.request.urlretrieve(video_url, "sample_video.mp4")

# Open the downloaded video
cap = cv2.VideoCapture("sample_video.mp4")

if not cap.isOpened():
    print("Error: Could not open the video.")
else:
    slow_factor = 0.5    # Scale down
    fast_factor = 2.0    # Scale up

    frame_count = 0
    max_frames = 3

    print("Processing Video...\n")

    while True:
        ret, frame = cap.read()

        if not ret:
            break

        # Resize for slow motion effect
        slow_frame = cv2.resize(frame, None, fx=slow_factor, fy=slow_factor)

        # Resize for fast motion effect
        fast_frame = cv2.resize(frame, None, fx=fast_factor, fy=fast_factor)

        print(f"Frame {frame_count + 1}")

        print("Original Frame")
        cv2_imshow(frame)

        print("Slow Motion Frame")
        cv2_imshow(slow_frame)

        print("Fast Motion Frame")
        cv2_imshow(fast_frame)

        print("-----")

        frame_count += 1

        if frame_count >= max_frames:
            break

    cap.release()

print("Video processing completed.")
```

Processing Video...

Frame 1

Original Frame



Slow Motion Frame



Fast Motion Frame



Frame 2

Original Frame



Slow Motion Frame



Fast Motion Frame

